Econ 527 HW3

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1 Speeding up VFI

This problem is the same form last problem set i.e. We seek to solve a simple representative agent neoclassical growth model. An agent solves

$$\begin{aligned} &\max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & k_{t+1} = Ak_t^{\alpha} + (1-\delta)k_t - c_t \end{aligned} \tag{1}$$

where $u(c) = \log(c)$, $k_t > 0$, and k_0 is given. The problem 1 can be written as

$$V(k) = \max_{k' \geq 0} \log\left(Ak^{\alpha} + (1-\delta)k - k'\right) + \beta V\left(k'\right) \tag{2}$$

We set the following parameters: $\beta = 0.96, A = 1, \delta = 1$, and $\alpha = 0.35$.

1.1 Howard Policy Iteration

To speed up Value Function Iteration, we implement the Howar Policy Iteration algorthim.

Algorithm 1 Howard Policy IterationAlgorithm

Require: Grid $\{k_i\}_{i=1}^N$, parameters β, α, δ, A, tolerance ϵ

Initialize $V^0(k_i) = 0$ for all i, set $n = 0$
repeat
  for each grid point k_i **do**
    Construct cubic spline interpolant $\tilde{V}^n$ from $\{k_i, V^n(k_i)\}$
    Solve $V^{n+1}(k_i) = \max_{k' \geq 0} \left\{ \log(Ak_i^\alpha - k') + \beta \tilde{V}^n(k') \right\}$
    if condition **then**
      do something
    else if other condition **then**
      do something else
    else
      default action
    end if
    Record optimizer $k'^*(k_i)$
  end for
  err $= \max_i |V^{n+1}(k_i) - V^n(k_i)|$
  $n \leftarrow n + 1$
until err $< \epsilon(1 - \beta)$

Ensure: Value function V^n and policy function k'^*

1.2 Endogeneous Grid Method

2 The Stationary Distribution of State Variables

2.1 Net Demand Using Interpolation

2.2 Net Demand Using Simulation